👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.NOS.2

Divide Multi-Digit Numbers Using an Algorithm

Lesson 2-6 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

Language Objective: I can explain my division using the words dividend, divisor, quotient, and remainder.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Algorithm · **Dividend** · **Divisor** · **Quotient** · **Remainder** · **Divide Multi-Digit Numbers** · **Long division** · **Bring down**

Level 3: try the blanks from memory first, then check the bank.

1 A step-by-step set of rules to solve a problem is called an _____.

2 In 84 ÷ 7, the total amount 84 being divided is the _____ .

3 In 84 ÷ 7, the number 7 that you divide by is the _____ .

4 The answer to a division problem is called the _____ .

5 When a number does not divide evenly, the amount left over is the _____ .

6 Long division repeats four steps: divide, multiply, subtract, _____ _____ .

7 The method that repeats divide, multiply, subtract, and bring down is called _____ .

8 After you subtract in long division, the next step is to _____ the next digit.

Write About the Math

How does multi-digit division help the school figure out how many pencils each classroom gets?
¿Cómo ayuda la división de varios dígitos a la escuela a averiguar cuántos lápices recibe cada salón?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Algorithm** *Algoritmo*
- ☐ **Dividend** *Dividendo*
- ☐ **Divisor** *Divisor*

- [ ] **Quotient**
  Cociente
- [ ] **Remainder**
  Residuo

Model: *Estimating with friendly numbers checks whether the quotient is reasonable.* — Students estimate using 1,200 ÷ 12 = 100 (or similar) to predict roughly 100+ bars per section, identifying 1,344 as the dividend and 12 as the divisor.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Each classroom gets ___ pencils because 1,680 ÷ 24 = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Algorithm
2. **Notes 2:** Dividend
3. **Notes 3:** Divisor
4. **Notes 4:** Quotient
5. **Notes 5:** Remainder
6. **Notes 6:** Divide Multi-Digit Numbers
7. **Notes 7:** Long division

8. **Notes 8:** Bring down
9. **Watch for this mistake:** *A common mistake in Divide Multi-Digit Numbers is skipping a placeholder zero in the quotient when a digit doesn't divide evenly, which shifts every digit after it out of place. For example, dividing 3,216 ÷ 3: 3 ÷ 3 = 1, but the next digit 2 doesn't divide by 3, so a 0 must be written before continuing — bring down the 1 to make 21 ÷ 3 = 7, then bring down 6 ÷ 3 = 2, giving the correct quotient 1,072. A student who skips writing that 0 and jumps straight to 21 ÷ 3 = 7 ends up with 172 instead of 1,072 — an answer off by a factor of ten. Always check: does a digit in your quotient go missing every time you 'can't divide'? If so, you dropped a placeholder zero.*
10. **Try It 1:** C. 704 — *7 × 704 = 4,928 exactly, so the quotient is 704 with no remainder.*
11. **Try It 2:** B. 671 R 6 — *9 × 671 = 6,039, and 6,045 − 6,039 = 6, so 6,045 ÷ 9 = 671 R 6.*